

# Model Question Paper-I with effect from 2023-24 (CBCS Scheme)

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## Third Semester B.E. Degree Examination Operating Systems

TIME: 03 Hours

Max. Marks: 100

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.

| Module -1 |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                             | *Bloom's Taxonomy Level | Marks        |            |          |    |   |    |   |    |    |   |   |    |   |   |    |    |   |   |   |    |   |
|-----------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|--------------|------------|----------|----|---|----|---|----|----|---|---|----|---|---|----|----|---|---|---|----|---|
| Q.01      | a            | Distinguish between the following terms.<br>(i) Multiprogramming and Multitasking<br>(ii) Multiprocessor System and Clustered System                                                                                                                                                                                                                                                                                                                        | L2                      | 6            |            |          |    |   |    |   |    |    |   |   |    |   |   |    |    |   |   |   |    |   |
|           | b            | Define operating Systems. Explain the dual-mode operating system with a neat diagram.                                                                                                                                                                                                                                                                                                                                                                       | L2                      | 7            |            |          |    |   |    |   |    |    |   |   |    |   |   |    |    |   |   |   |    |   |
|           | c            | With a neat diagram, explain the concept of the virtual machine.                                                                                                                                                                                                                                                                                                                                                                                            | L2                      | 7            |            |          |    |   |    |   |    |    |   |   |    |   |   |    |    |   |   |   |    |   |
| OR        |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                         |              |            |          |    |   |    |   |    |    |   |   |    |   |   |    |    |   |   |   |    |   |
| Q.02      | a            | Explain the layered approach of operating system structure with a supporting diagram.                                                                                                                                                                                                                                                                                                                                                                       | L2                      | 6            |            |          |    |   |    |   |    |    |   |   |    |   |   |    |    |   |   |   |    |   |
|           | b            | What are system calls? Briefly point out its types with illustrations.                                                                                                                                                                                                                                                                                                                                                                                      | L2                      | 6            |            |          |    |   |    |   |    |    |   |   |    |   |   |    |    |   |   |   |    |   |
|           | c            | Explain the services of the operating system that are helpful for the user and the system.                                                                                                                                                                                                                                                                                                                                                                  | L2                      | 8            |            |          |    |   |    |   |    |    |   |   |    |   |   |    |    |   |   |   |    |   |
| Module-2  |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                         |              |            |          |    |   |    |   |    |    |   |   |    |   |   |    |    |   |   |   |    |   |
| Q. 03     | a            | With a neat diagram, explain the states of a process with a transition diagram and process control block.                                                                                                                                                                                                                                                                                                                                                   | L2                      | 6            |            |          |    |   |    |   |    |    |   |   |    |   |   |    |    |   |   |   |    |   |
|           | b            | What is inter-process communication? Discuss message passing and the shared memory concept of IPC.                                                                                                                                                                                                                                                                                                                                                          | L2                      | 6            |            |          |    |   |    |   |    |    |   |   |    |   |   |    |    |   |   |   |    |   |
|           | c            | Calculate average waiting and turnaround times by drawing the Gantt chart using FCFS and RR (q=2ms). <table><tr><td>Processes</td><td>Arrival Time</td><td>Burst Time</td></tr><tr><td>P1</td><td>0</td><td>9</td></tr><tr><td>P2</td><td>1</td><td>4</td></tr><tr><td>P3</td><td>2</td><td>9</td></tr><tr><td>P4</td><td>3</td><td>5</td></tr></table>                                                                                                     | Processes               | Arrival Time | Burst Time | P1       | 0  | 9 | P2 | 1 | 4  | P3 | 2 | 9 | P4 | 3 | 5 | L3 | 8  |   |   |   |    |   |
| Processes | Arrival Time | Burst Time                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                         |              |            |          |    |   |    |   |    |    |   |   |    |   |   |    |    |   |   |   |    |   |
| P1        | 0            | 9                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                         |              |            |          |    |   |    |   |    |    |   |   |    |   |   |    |    |   |   |   |    |   |
| P2        | 1            | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                         |              |            |          |    |   |    |   |    |    |   |   |    |   |   |    |    |   |   |   |    |   |
| P3        | 2            | 9                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                         |              |            |          |    |   |    |   |    |    |   |   |    |   |   |    |    |   |   |   |    |   |
| P4        | 3            | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                         |              |            |          |    |   |    |   |    |    |   |   |    |   |   |    |    |   |   |   |    |   |
| OR        |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                         |              |            |          |    |   |    |   |    |    |   |   |    |   |   |    |    |   |   |   |    |   |
| Q.04      | a            | Discuss in detail the multithreading model, its advantages and disadvantages with suitable illustration.                                                                                                                                                                                                                                                                                                                                                    | L2                      | 7            |            |          |    |   |    |   |    |    |   |   |    |   |   |    |    |   |   |   |    |   |
|           | b            | Explain five different scheduling criteria used in the computing scheduling mechanism.                                                                                                                                                                                                                                                                                                                                                                      | L2                      | 5            |            |          |    |   |    |   |    |    |   |   |    |   |   |    |    |   |   |   |    |   |
|           | c            | Calculate the average waiting time and the average turnaround time by drawing the Gantt chart using SRTF and the Priority scheduling algorithm. <table><tr><td>Processes</td><td>Arrival Time</td><td>Burst Time</td><td>Priority</td></tr><tr><td>P1</td><td>0</td><td>8</td><td>3</td></tr><tr><td>P2</td><td>1</td><td>4</td><td>2</td></tr><tr><td>P3</td><td>2</td><td>9</td><td>1</td></tr><tr><td>P4</td><td>3</td><td>5</td><td>4</td></tr></table> | Processes               | Arrival Time | Burst Time | Priority | P1 | 0 | 8  | 3 | P2 | 1  | 4 | 2 | P3 | 2 | 9 | 1  | P4 | 3 | 5 | 4 | L3 | 8 |
| Processes | Arrival Time | Burst Time                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Priority                |              |            |          |    |   |    |   |    |    |   |   |    |   |   |    |    |   |   |   |    |   |
| P1        | 0            | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3                       |              |            |          |    |   |    |   |    |    |   |   |    |   |   |    |    |   |   |   |    |   |
| P2        | 1            | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2                       |              |            |          |    |   |    |   |    |    |   |   |    |   |   |    |    |   |   |   |    |   |
| P3        | 2            | 9                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1                       |              |            |          |    |   |    |   |    |    |   |   |    |   |   |    |    |   |   |   |    |   |
| P4        | 3            | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4                       |              |            |          |    |   |    |   |    |    |   |   |    |   |   |    |    |   |   |   |    |   |

| Module-3       |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |   |     |   |   |   |           |   |   |    |            |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
|----------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|-----|---|---|---|-----------|---|---|----|------------|--|--|--|-----|--|--|--|-----------|--|--|--|--|---|---|---|---|---|---|---|---|---|---|---|---|----------------|---|---|---|---|---|---|---|---|---|---|---|---|----------------|---|---|---|---|---|---|---|---|--|--|--|--|----------------|---|---|---|---|---|---|---|---|--|--|--|--|----------------|---|---|---|---|---|---|---|---|--|--|--|--|----------------|---|---|---|---|---|---|---|---|--|--|--|--|----|
| Q. 05          | a          | Define deadlock. What are the necessary conditions for deadlock to occur?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |   |     |   |   |   |           |   |   | L2 | 5          |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
|                | b          | Illustrate Peterson's solution for the critical section problem.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |   |     |   |   |   |           |   |   | L2 | 6          |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
|                | c          | Consider the following snapshot of the system:<br><table border="1"><thead><tr><th></th><th colspan="4">Allocation</th><th colspan="4">Max</th><th colspan="4">Available</th></tr><tr><th></th><th>A</th><th>B</th><th>C</th><th>D</th><th>A</th><th>B</th><th>C</th><th>D</th><th>A</th><th>B</th><th>C</th><th>D</th></tr></thead><tbody><tr><td>P<sub>0</sub></td><td>0</td><td>0</td><td>1</td><td>2</td><td>0</td><td>0</td><td>1</td><td>2</td><td>1</td><td>5</td><td>2</td><td>0</td></tr><tr><td>P<sub>1</sub></td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>7</td><td>5</td><td>0</td><td></td><td></td><td></td><td></td></tr><tr><td>P<sub>2</sub></td><td>1</td><td>3</td><td>5</td><td>4</td><td>2</td><td>3</td><td>5</td><td>6</td><td></td><td></td><td></td><td></td></tr><tr><td>P<sub>3</sub></td><td>0</td><td>6</td><td>3</td><td>2</td><td>0</td><td>6</td><td>5</td><td>2</td><td></td><td></td><td></td><td></td></tr><tr><td>P<sub>4</sub></td><td>0</td><td>0</td><td>1</td><td>4</td><td>0</td><td>6</td><td>5</td><td>6</td><td></td><td></td><td></td><td></td></tr></tbody></table><br>Answer the following questions using Banker's algorithm:<br>a. What is the content of the need matrix?<br>b. Is the system in a safe state? If yes, mention the safe sequence.<br>If a request from process P1 arrives for (0,4,2,0) can the request be granted immediately? |   |   |     |   |   |   |           |   |   |    | Allocation |  |  |  | Max |  |  |  | Available |  |  |  |  | A | B | C | D | A | B | C | D | A | B | C | D | P <sub>0</sub> | 0 | 0 | 1 | 2 | 0 | 0 | 1 | 2 | 1 | 5 | 2 | 0 | P <sub>1</sub> | 1 | 0 | 0 | 0 | 1 | 7 | 5 | 0 |  |  |  |  | P <sub>2</sub> | 1 | 3 | 5 | 4 | 2 | 3 | 5 | 6 |  |  |  |  | P <sub>3</sub> | 0 | 6 | 3 | 2 | 0 | 6 | 5 | 2 |  |  |  |  | P <sub>4</sub> | 0 | 0 | 1 | 4 | 0 | 6 | 5 | 6 |  |  |  |  | L3 |
|                | Allocation |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |   | Max |   |   |   | Available |   |   |    |            |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
|                | A          | B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | C | D | A   | B | C | D | A         | B | C | D  |            |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
| P <sub>0</sub> | 0          | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1 | 2 | 0   | 0 | 1 | 2 | 1         | 5 | 2 | 0  |            |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
| P <sub>1</sub> | 1          | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0 | 0 | 1   | 7 | 5 | 0 |           |   |   |    |            |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
| P <sub>2</sub> | 1          | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 5 | 4 | 2   | 3 | 5 | 6 |           |   |   |    |            |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
| P <sub>3</sub> | 0          | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 3 | 2 | 0   | 6 | 5 | 2 |           |   |   |    |            |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
| P <sub>4</sub> | 0          | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1 | 4 | 0   | 6 | 5 | 6 |           |   |   |    |            |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
| OR             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |   |     |   |   |   |           |   |   |    |            |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
| Q. 06          | a          | Explain different methods to recover from deadlocks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |   |     |   |   |   |           |   |   | L2 | 6          |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
|                | b          | What is a resource allocation graph? Consider an example to explain how it is very useful in describing a deadly embrace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |   |     |   |   |   |           |   |   | L2 | 6          |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
|                | c          | What is a semaphore? State a Dining Philosopher problem gives a solution using semaphore.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |   |     |   |   |   |           |   |   | L2 | 8          |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
| Module-4       |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |   |     |   |   |   |           |   |   |    |            |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
| Q. 07          | a          | What is TLB? Explain TLB in detail with a paging system with a neat diagram.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |   |     |   |   |   |           |   |   | L2 | 6          |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
|                | b          | With the help of a neat diagram, explain the various steps of address binding.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |   |     |   |   |   |           |   |   | L2 | 6          |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
|                | c          | Consider the page reference string: 1,0,7,1,0,2,1,2,3,0,3,2,4,0,3,6,2,1 for a memory with three frames. Determine the number of page faults using the FIFO, Optimal, and LRU replacement algorithms. Which algorithm is most efficient?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |   |     |   |   |   |           |   |   | L3 | 8          |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
| OR             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |   |     |   |   |   |           |   |   |    |            |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
| Q. 08          | a          | What is demand paging? Explain the steps in handling page faults using the appropriate diagram.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |   |     |   |   |   |           |   |   | L2 | 6          |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
|                | b          | What is segmentation? Explain the basic method of segmentation with an example.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |   |     |   |   |   |           |   |   | L2 | 6          |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
|                | c          | Discuss the structure of the page table with a suitable diagram.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |   |     |   |   |   |           |   |   | L2 | 8          |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
| Module-5       |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |   |     |   |   |   |           |   |   |    |            |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
| Q. 09          | a          | What is a file? What are its attributes? Explain file operations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |   |     |   |   |   |           |   |   | L2 | 6          |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
|                | b          | Explain in detail about various file operations in a file system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |   |     |   |   |   |           |   |   | L2 | 6          |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
|                | c          | Discuss various directory structures with neat diagrams.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |   |     |   |   |   |           |   |   | L2 | 8          |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
| OR             |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |   |     |   |   |   |           |   |   |    |            |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
| Q. 10          | a          | Explain contiguous and linked disk space allocation methods.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |   |     |   |   |   |           |   |   | L2 | 6          |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
|                | b          | Explain the access matrix method of system protection with the domain as objects and its implementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |   |     |   |   |   |           |   |   | L2 | 6          |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |
|                | c          | Given the following sequences 95,180,34,119,11,123,62,64 with the track 50 and ending track 199. What is the total disk travelled by the disk arm using FCFS, SSTF, LOOK and CLOOK algorithm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |   |     |   |   |   |           |   |   | L3 | 8          |  |  |  |     |  |  |  |           |  |  |  |  |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |   |   |   |   |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |                |   |   |   |   |   |   |   |   |  |  |  |  |    |

\*Bloom's Taxonomy Level: Indicate as L1, L2, L3, L4, etc. It is also desirable to indicate the COs and POs to be attained by every bit of question.